

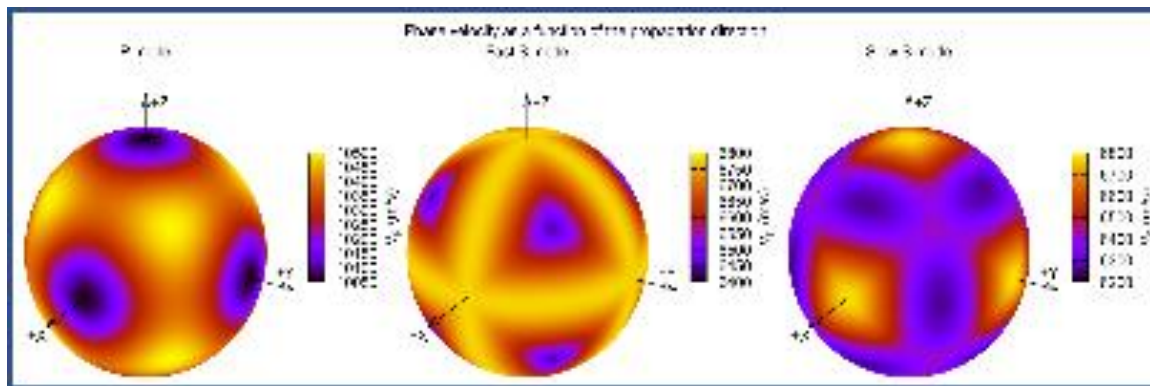
Mechanical Response of Materials: Pressure and Elasticity

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MSSC2024 London 20/09/2022

Crystal23 Tutorials and theory

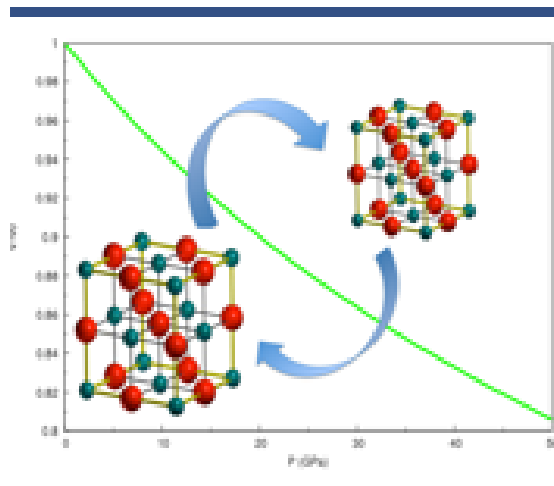
CRYSTAL23 Manual pages 282-303



Elastic, Piezoelectric and Photoelastic Tensors

A. Erba and E. Albanese

http://www.theochem.unito.it/crystal_tuto/ms_sc2016_cd/tutorials17/elapiezo/photo/photo.html



Equation of State

A. Mahmoud and A. Erba

http://www.theochem.unito.it/crystal_tuto/ms_sc2013_cd/tutorials/eos/eos_tut.html

MOTIVATION and IMPORTANCE

PCCP

PAPER

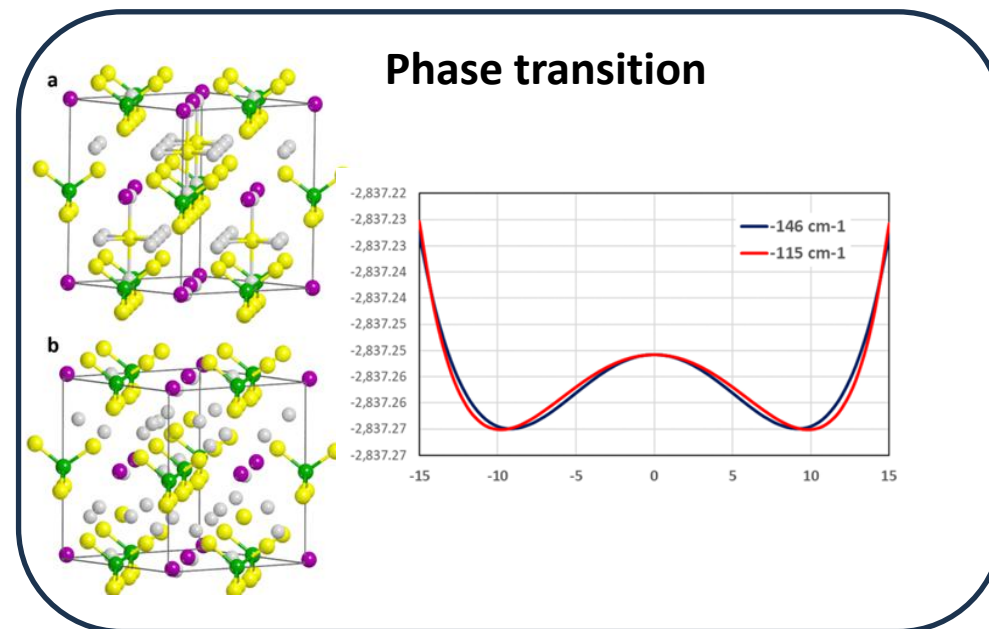
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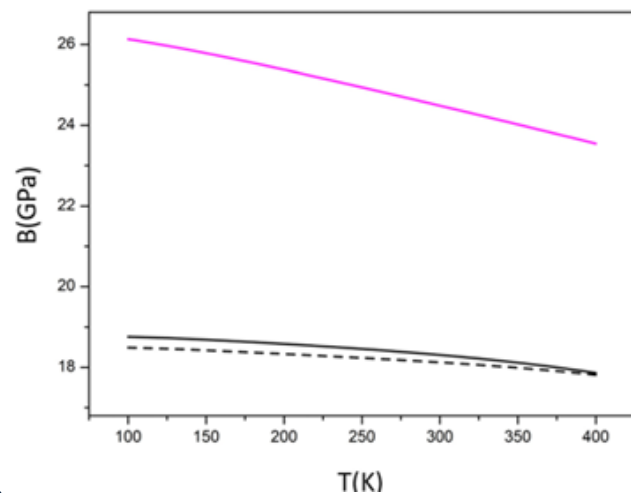
Cite this: *Phys. Chem. Chem. Phys.*,
2022, 24, 22978

From symmetry breaking in the bulk to phase transitions at the surface: a quantum-mechanical exploration of $\text{Li}_6\text{PS}_5\text{Cl}$ argyrodite superionic conductor†

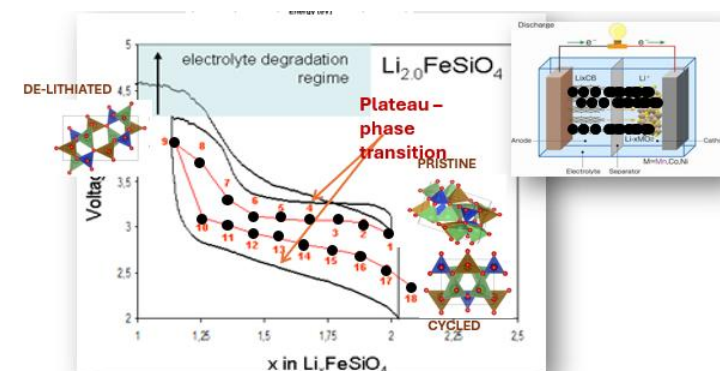
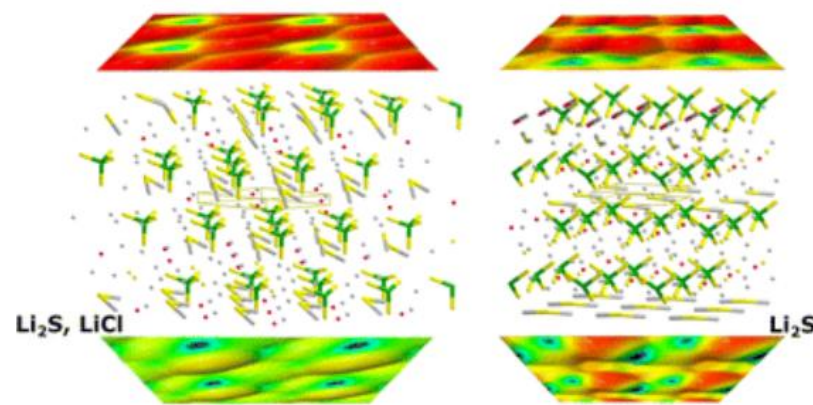
Maddalena D'Amore,^a Loredana Edith Daga,^a Riccardo Rocca,^{ab}
Mauro Francesco Sgroi,^b Naiara Leticia Marana,^a Silvia Maria Casassa,^a
Lorenzo Maschio^a and Anna Maria Ferrari^a



Bulk modulus



Surface phase transitions





LAYOUT

- Phase Transitions

Phase Diagrams

Equation of States (EOS)

Bulk modulus

Compressibility

EOS

- Elastic Constants

ELASTCON

THERMODYNAMICS OF SOLUTIONS

1 component system (for a given phase):

The thermodynamic state is determined uniquely if any two of its thermodynamical properties (variables) are known.

Variables are:

- . Temperature (T)**
- . Pressure (P)**
- . Volume (V)**
- . Enthalpy (H)**
- . Entropy (S)**
- . Free Energy (G)**

G: Constant Pressure

$$\mathbf{G = H - TS}$$

A: Constant Volume

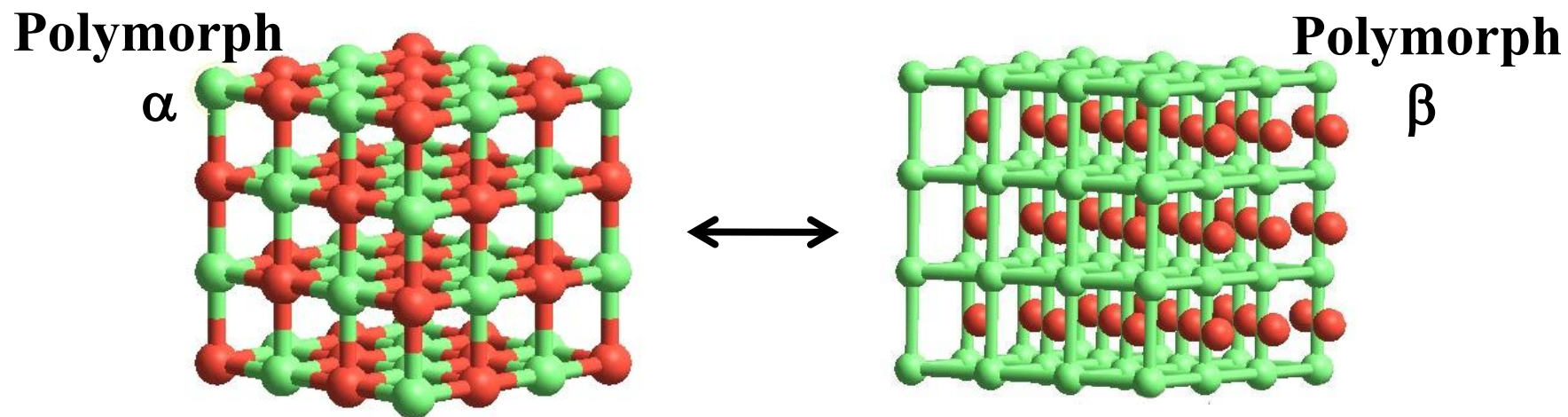
$$\mathbf{A = U - TS}$$

$$\mathbf{H = U + PV ; U = \text{Internal Energy}}$$

THERMODYNAMICS OF SOLUTIONS

$$G = U + PV - TS$$

The stable polymorph at given T and P is the one with the lowest
Gibb's free energy (G)



$$G^{\alpha} < G^{\beta}; G^{\alpha}$$

$$G^{\alpha} > G^{\beta}; G^{\beta}$$

$$G^{\alpha} = G^{\beta}; \text{Equilibrium}$$

GIBB'S FREE ENERGY and PHASE TRANSITIONS

$$G^{\alpha} = G^{\beta} = \Delta G = 0$$



$$\Delta H = T\Delta S$$

At low temperatures the phase with smaller enthalpy is the stable phase.

At low temperatures:

$$G = H = U + PV$$

U = Internal energy = $E(\text{SCF})$

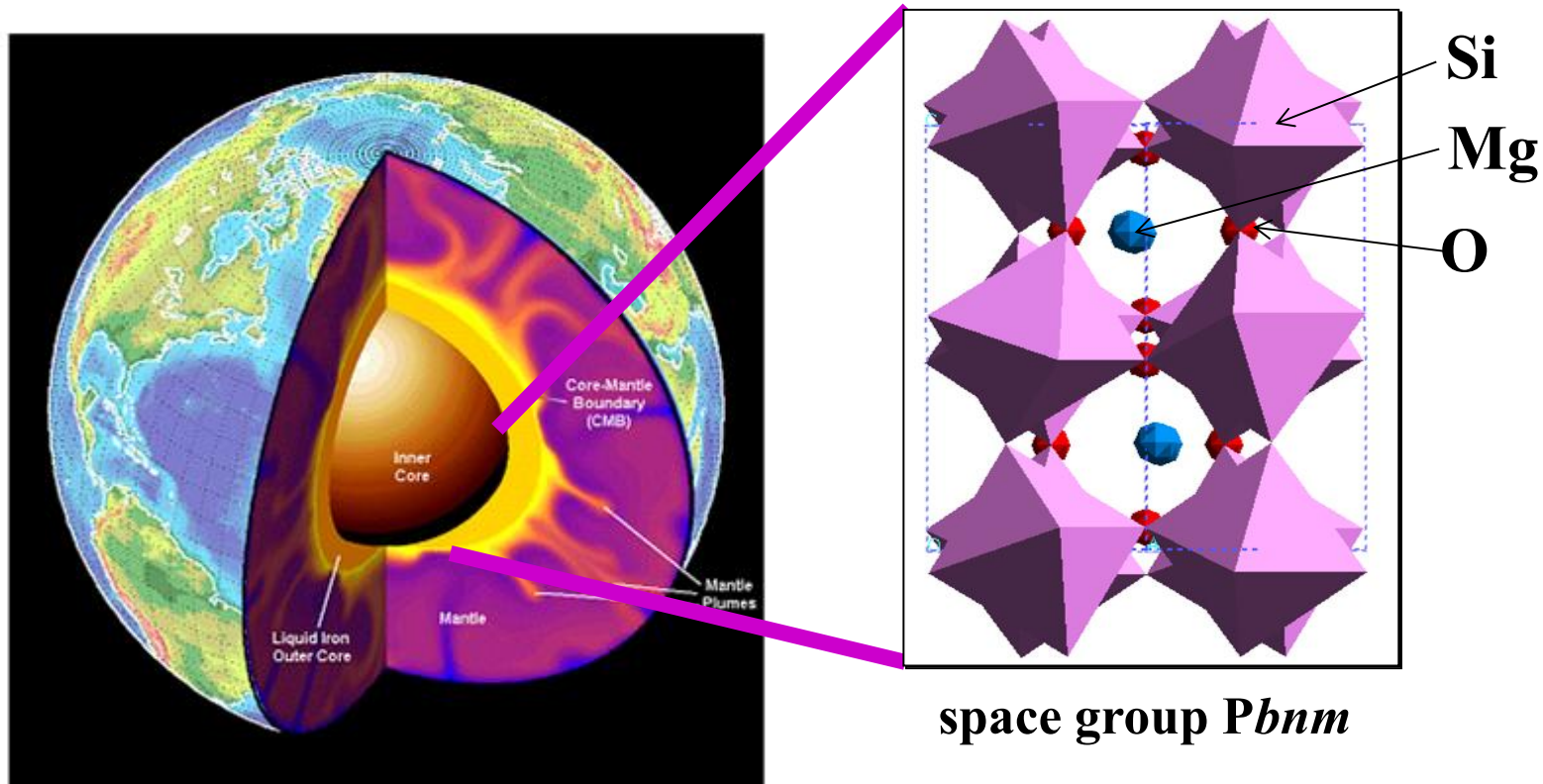
In quantum mechanics we often work at 0 K
 $\Rightarrow$ CRYSTAL ($T=0$ K)
WITH PHONON (DISPERSION) $\rightarrow$ TEMPERATURE EFFECTS

WHY STUDY MGO?

The perovskite structure of MgSiO_3 exists in the Earth's lower mantle, formed under high-pressure conditions according to:

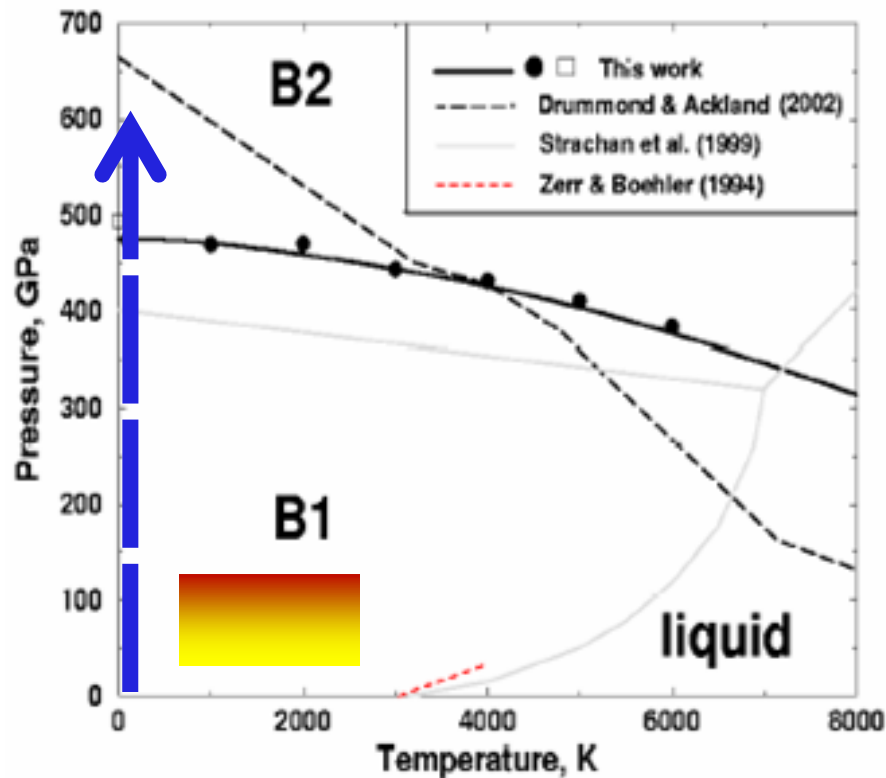


$$T_{\text{l-mantle}} = 800\text{--}2700 \text{ K}; P_{\text{l-mantle}} = 25\text{--}137 \text{ GPa}$$

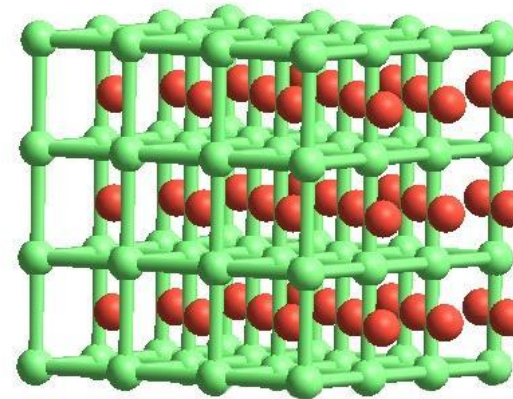


MgO AS A COMPUTATIONAL EXAMPLE

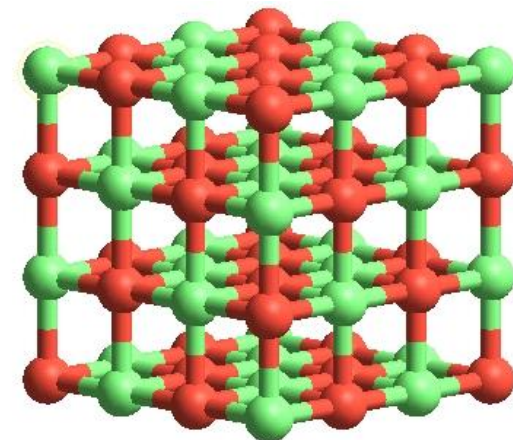
$$G = H = U + PV$$



Oganov et al., J.Chem.Phys. **118**, 10174 (2003)



B2
(*NiAs structure*)

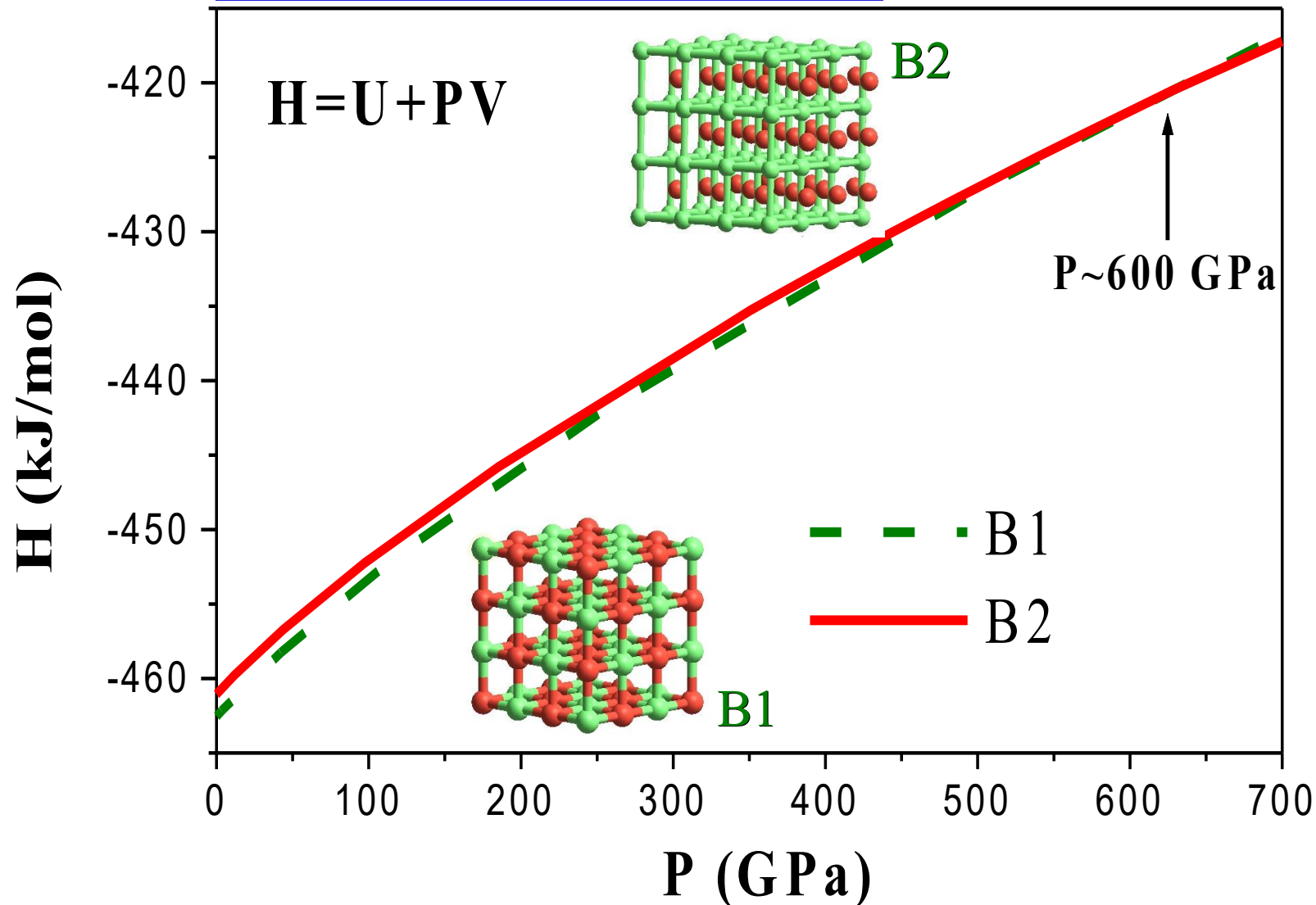


B1
(*NaCl structure*)

1. PRESSURE (P) IS KNOWN

$$H = U + PV$$

Keep P fixed in optimisation



OPTGEOM
EXTPRESS
0.001
END
NOTE!
hartree/bohr³

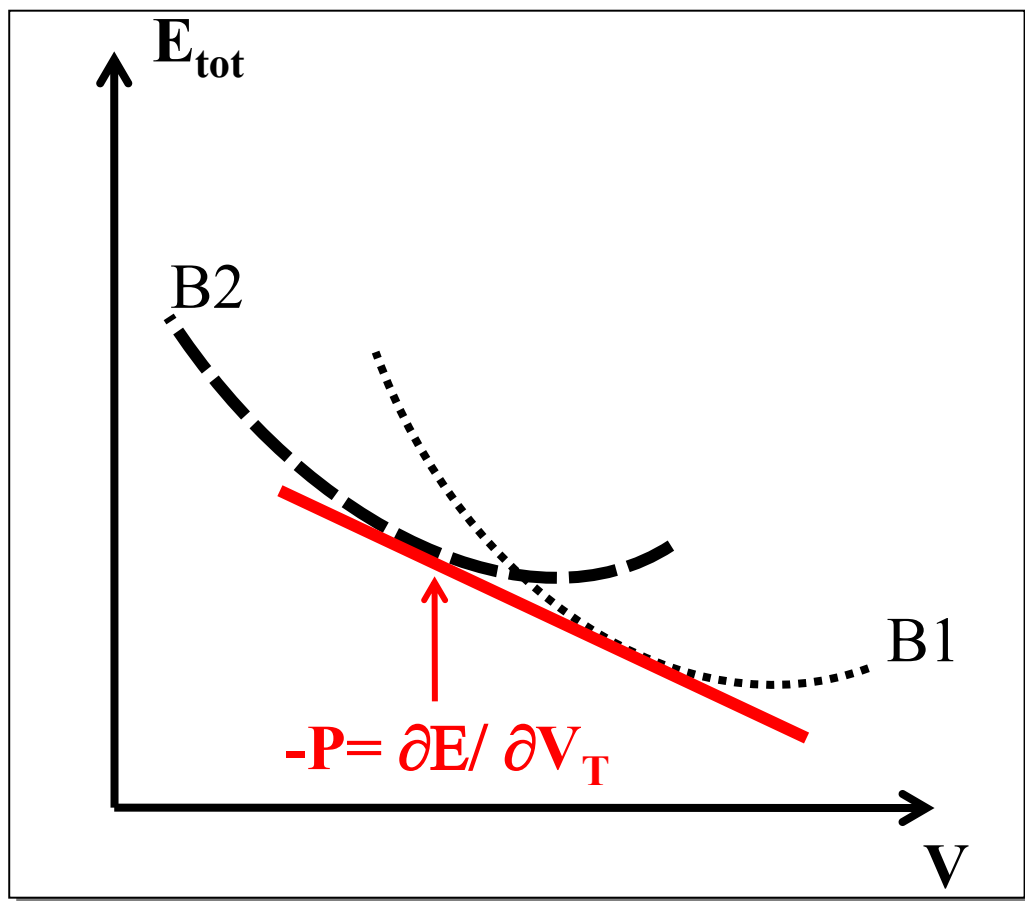
- Perform geometry optimisation for both structures at constant pressure
- Plot enthalpy over pressure
- Identify at what pressure the curves cross

2. VOLUME (V) IS KNOWN

From thermodynamics we know:

$$\left(\frac{\partial U}{\partial V} \right)_S = -P$$

Keep V fixed in optimisation



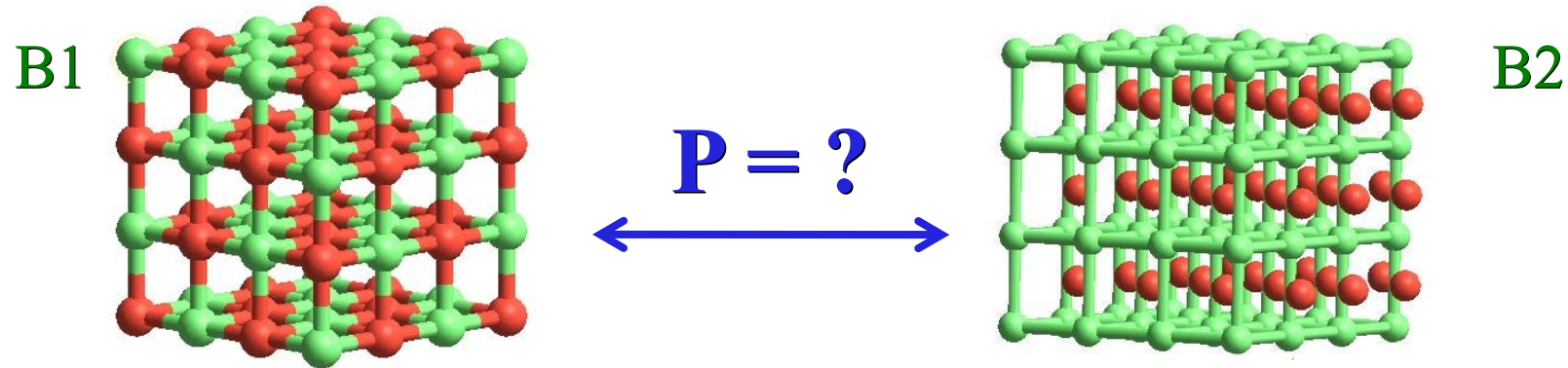
The pressure that corresponds to a given volume can be obtained from the 1st derivative of $\partial E / \partial V$

USE KEYWORD:

```
OPTGEOM
CVOLOPT
END
or
OPTGEOM
EOS
RANGE
0.90 1.05 10
END
END
```

Effect of Hamiltonian on P_T

The MgO case

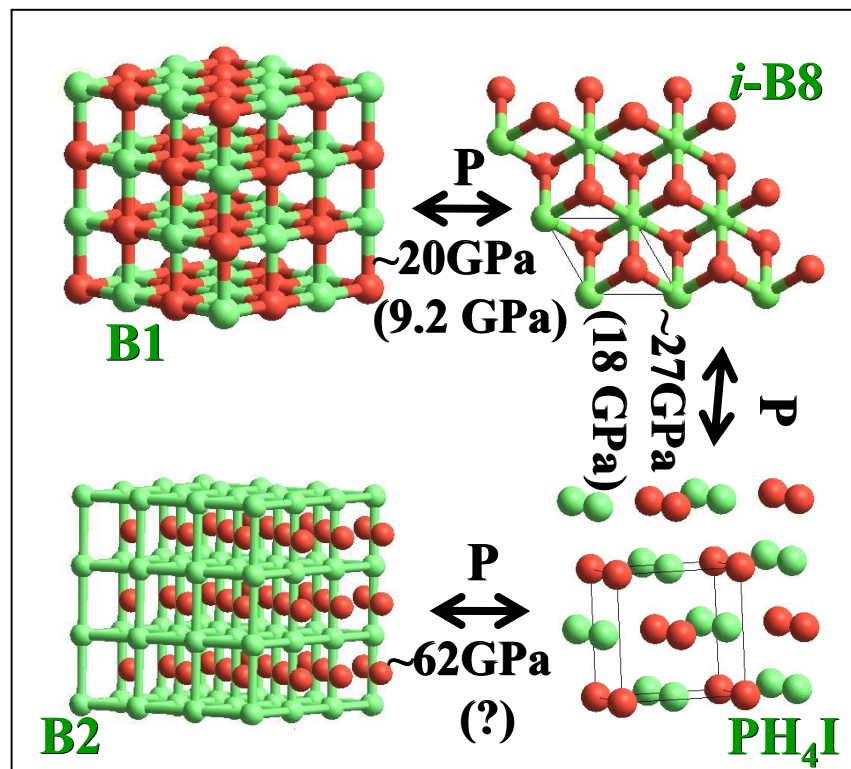


Hamiltonian	P_T (GPa)
HF	712
LDA	512
BLYP	478
PBE	428
PW91	418
Expt.	?

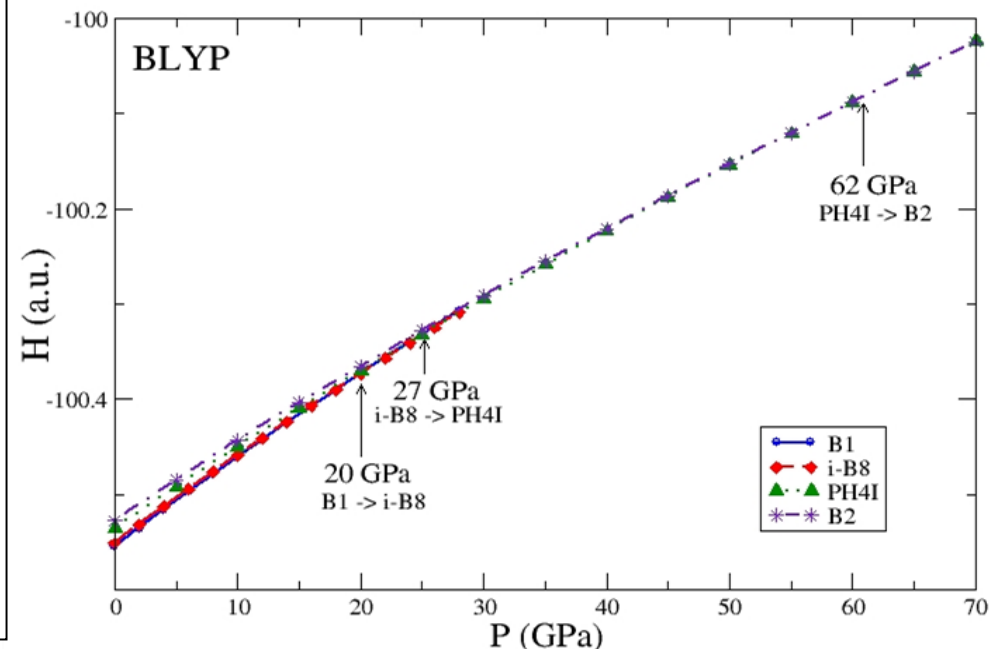
A blue arrow points from the LDA value (512 GPa) to a large question mark on the right.

PHASE TRANSITIONS FROM 1ST-PRINCIPLES CALCUALTIONS

The BaO case



Expt. in parenthesis:
Liu et al. JGR, 77 (1972) 4934



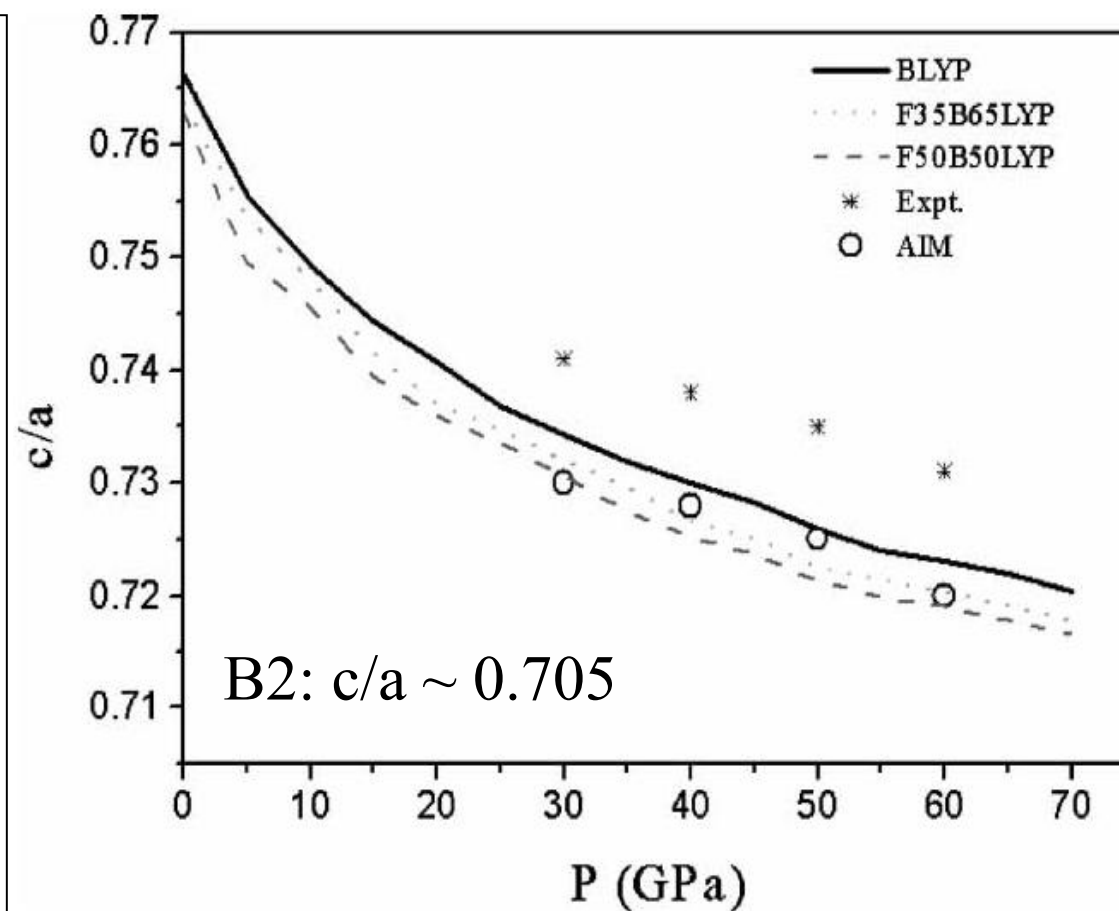
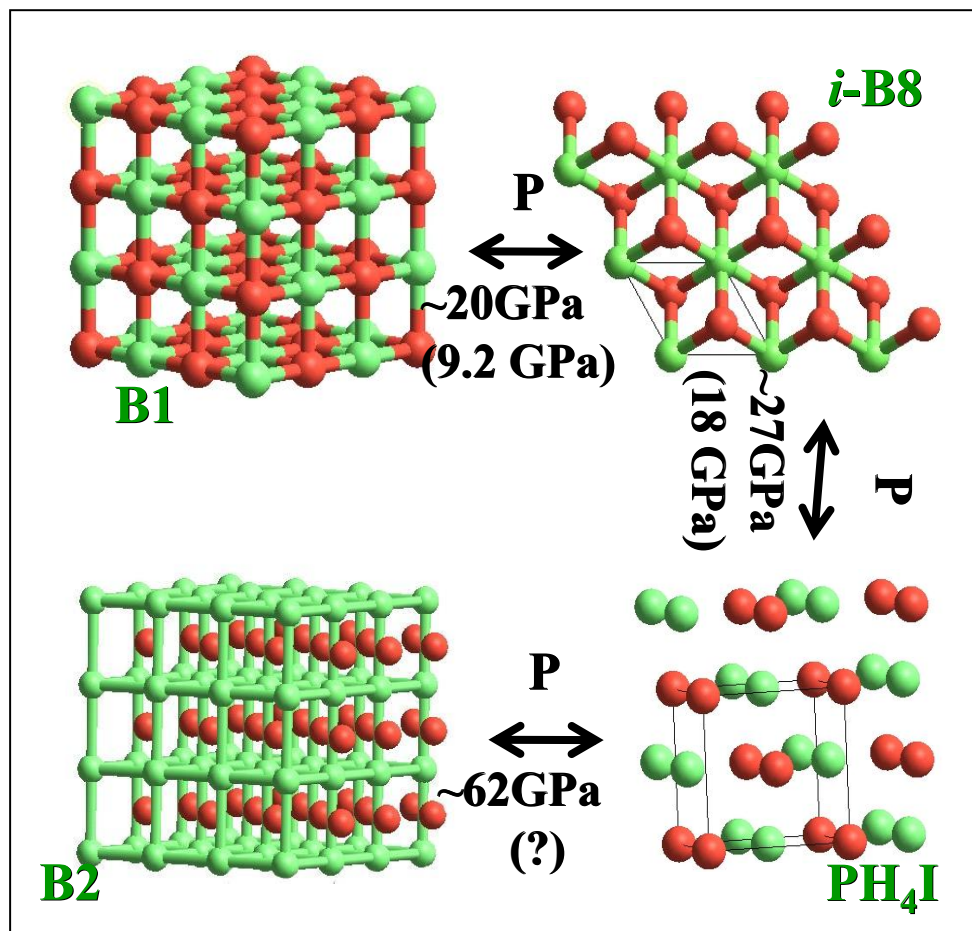
Alfredsson et al. *Mol. Sim.* (2004)

To determine phase transitions we need to:
Optimise the geometries for all the possible crystal structures at various pressures.

Experimentally B2 never observed, why is it possible to see theoretically?

PHASE TRANSITIONS FROM 1ST-PRINCIPLES CALCUALTIONS

The BaO case



Alfredsson et al. *Mol. Sim.* (2004)

Experimentally PH₄I slowly transforms into B2
Experiments done up to 60 GPa

1st and 2nd ORDER PHASE TRANSITIONS

- **1st order:** involve a latent heat, i.e. as system either absorbs or releases a fixed amount of energy $\Rightarrow$ “mixed-phase regimes”



Constant
temperature

- **2nd order:** continuous phase transition. (no associated latent heat)
examples: magnetic transitions, liquid crystals etc.

Bulk Moduli and EQUATION OF STATEs (EOS)

Thermodynamic state of a system is usually defined by:

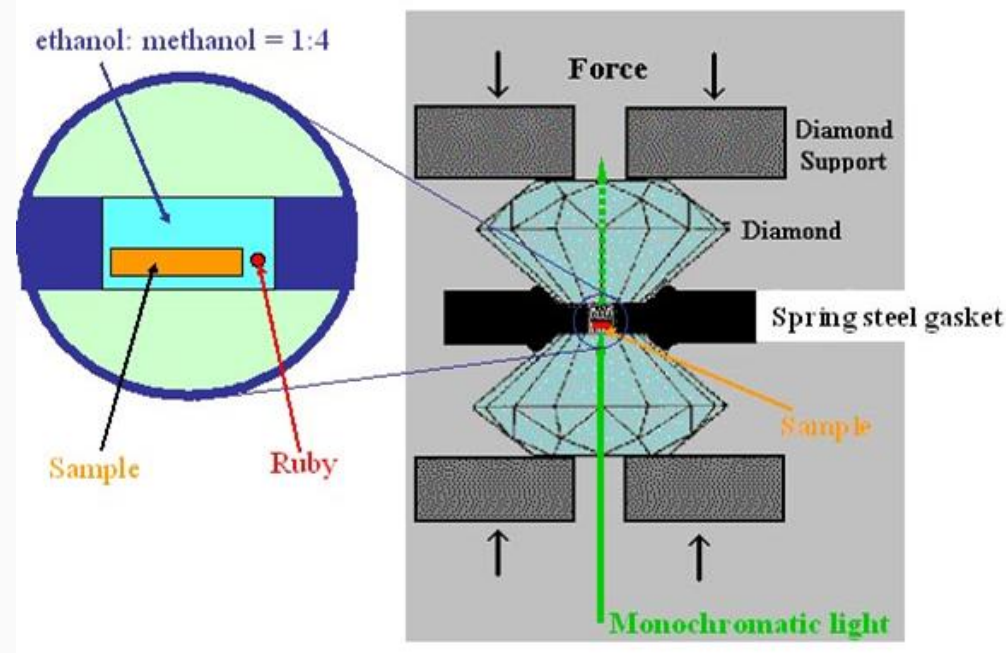
P, T, V linked by the Equation of State (EOS)

The best known EOS is the one for ideal gases:

$$PV = nRT$$

We are concerned about solids.

Applying an
hydrostatic P
to a solid sample



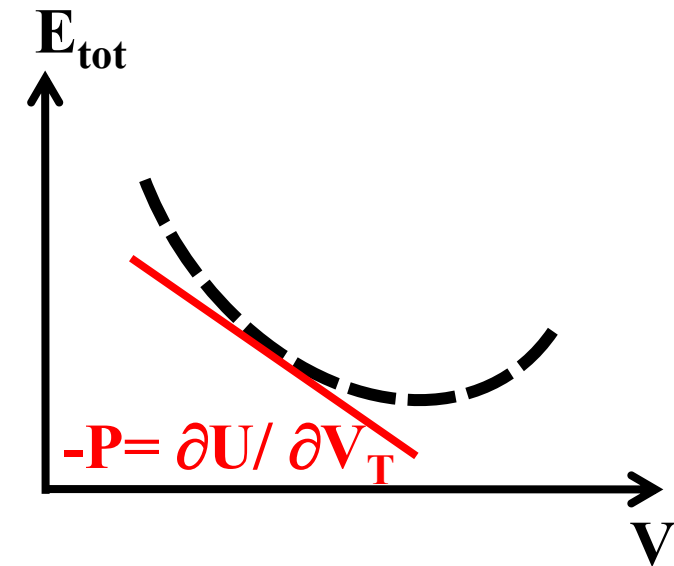
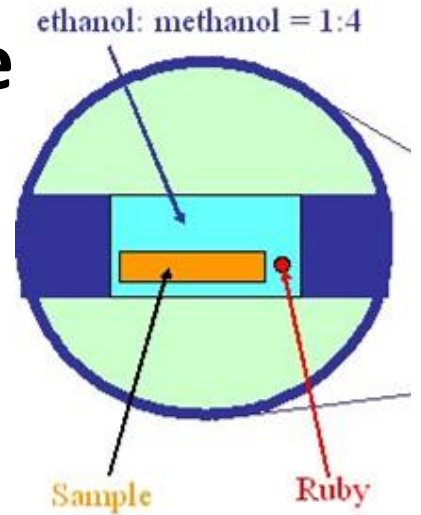
Bulk Modulus and EQUATION OF STATEs (EOS)

For a solid the simplest equation of state is given by the definition the bulk modulus (K)*

$$K = -V \left(\frac{\partial P}{\partial V} \right)_T$$

using $-P = \partial U / \partial V$

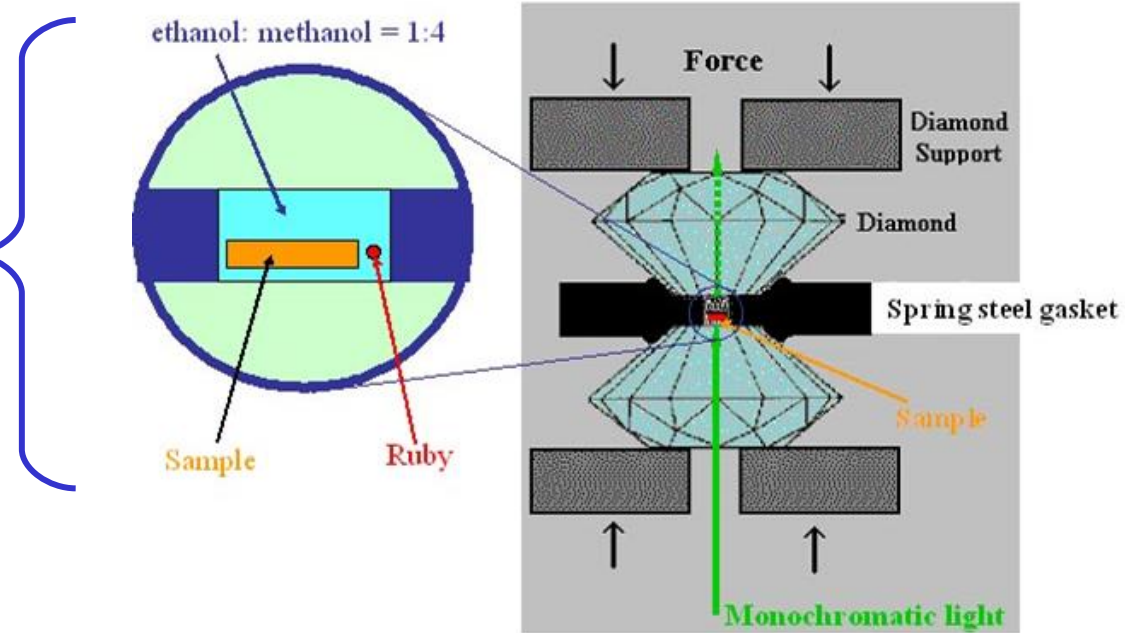
$$K = V \left(\frac{\partial^2 U}{\partial V^2} \right)_T$$



*Bulk modulus is often and in CRYSTAL17 manual abbreviated as B

Bulk Modulus and EQUATION OF STATEs (EOS)

Applying an
hydrostatic P
to a solid sample



The bulk modulus of a sample is a measure of its compressibility!

Example: How much pressure shall we apply if we want to decrease the volume of MgO (bulk modulus of 162 GPa) with 1%?

Bulk Modulus and EQUATION OF STATEs (EOS)

$$K = -V \left(\frac{\partial P}{\partial V} \right)_T$$

rewrite:

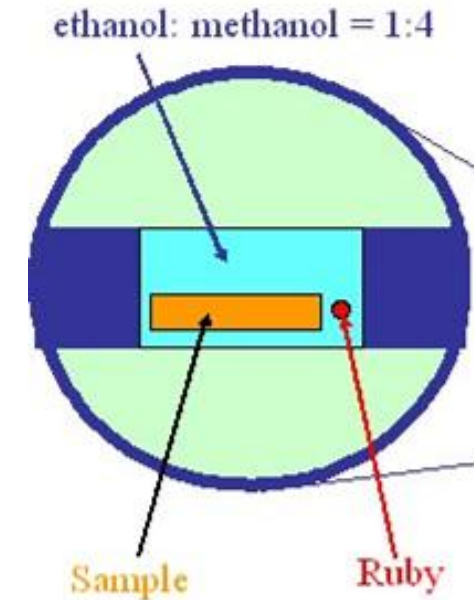
$$\frac{dV}{V} = -\frac{dP}{K}$$

integrate:

$$\ln \frac{V}{V_0} = \frac{P_0 - P}{K}$$

rewrite:

$$V = V_0 e^{\left(-\frac{P}{K} \right)}$$



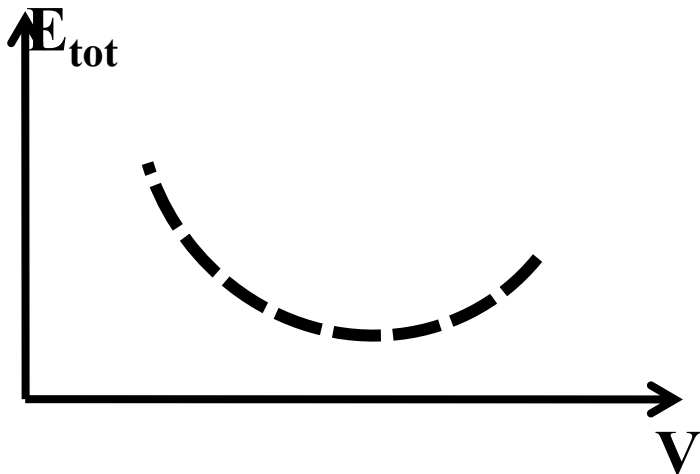
**This EOS suggest that K is the same also for high pressures.
However, we know that K becomes more incompressible for high pressures.
This EOS is not good for high pressures!!!!**

EQUATION OF STATEs (EOS)

- Murnaghan (Murnaghan F.D., (1967) Finite deformation of an elastic Solid, Dover, New York)
- Birch-Murnaghan (Birch F., (1978), JGR, 83, 1257)
- 2nd order Birch-Murnaghan (Bullen K.E., (1975) The Earth's density, Chapman and Hall, London, UK)
- 3rd order Birch-Murnaghan (Poirer J.P. (2000) Earth's interior, Cambridge Univ. Press, UK)
- Hencky finite strain (Poirer J.P. (2000) Earth's interior, Cambridge Univ. Press, UK)

See **CRYSTAL**
manual p. 282
EOS
RANGE
and the
tutorial

fit e.g. Murnaghan's equation $\Rightarrow V_0$ and K_0



+++++ FITTING USING ALL POINTS +++++

EQUATION OF STATE	VOL(A^3)	E(AU)	BM(GPA)	BM PRIME
MURNAGHAN 1944	116.7247	-1319.75250331	43.81	3.78
BIRCH-MURNAGHAN 1947	116.7202	-1319.75250331	43.83	3.88
POIRIER-TARANTOLA 1998	116.7172	-1319.75250341	43.85	3.95
VINET 1987	116.7186	-1319.75250334	43.84	3.92

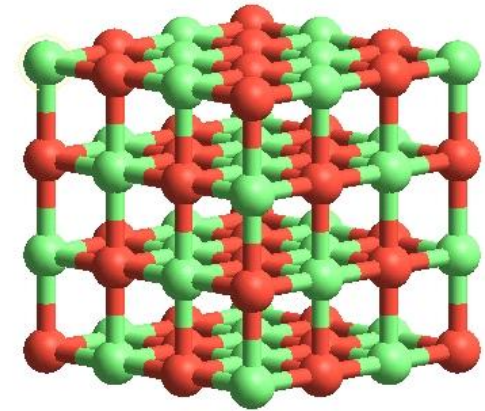
MgO (B1) structure BLYP Hamiltonian

EOS	V ($\text{\AA}^3$)	K (GPa)	K'
Fit parabola	19.51	143.50	-
Murnaghan	19.49	151.77	3.93
2 nd order B-M	19.48	152.29	-
3 rd order B-M	19.49	151.13	4.13
3 rd order Lagrangian	19.49	147.88	5.28
Davis and Gordon	19.49	150.67	4.28
Bardeen	19.49	151.04	4.16
Slater	19.49	150.87	4.21
Brennan and Stacey	19.49	150.76	4.25

EQUATION OF STATEs (EOS)

MgO (B1) structure with different Hamiltonians

Hamiltonian	V (Å ³)	K (GPa)
HF	18.4	182
LDA	18.0	181
BLYP	19.5	153
PBE	19.1	160
PW91	19.1	159
Expt.	18.5	162



B1

Habas et al. J. Phys.: Condens. Matter, 10 (1998) 6897

KEYWORD IN CRYSTAL

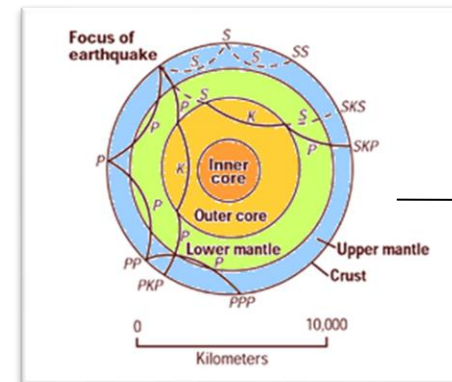
EOS

....

END

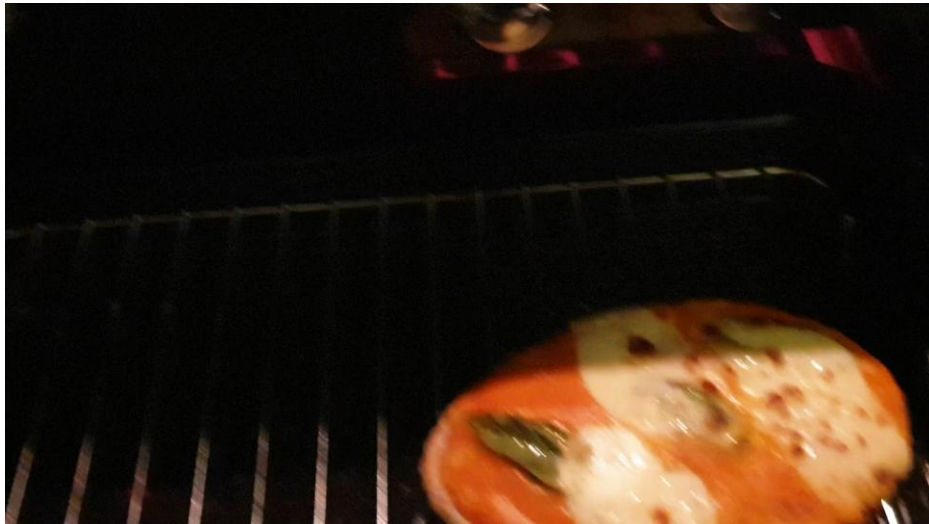
Elastic Constants

Information about materials properties
Brittle versus chewy

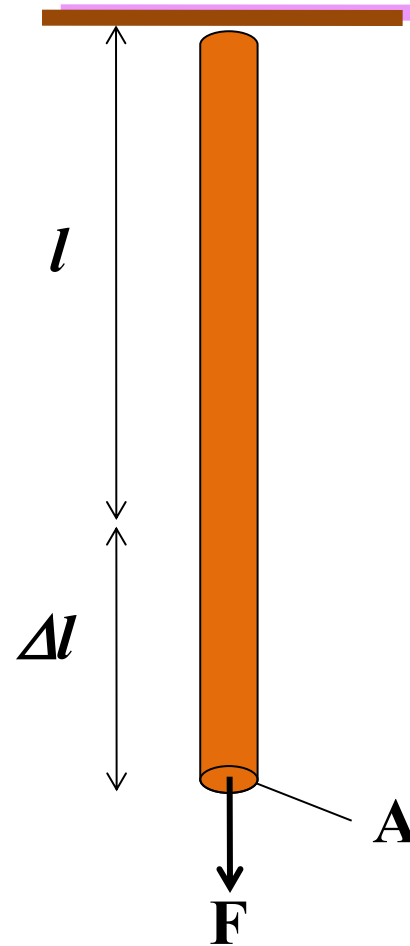
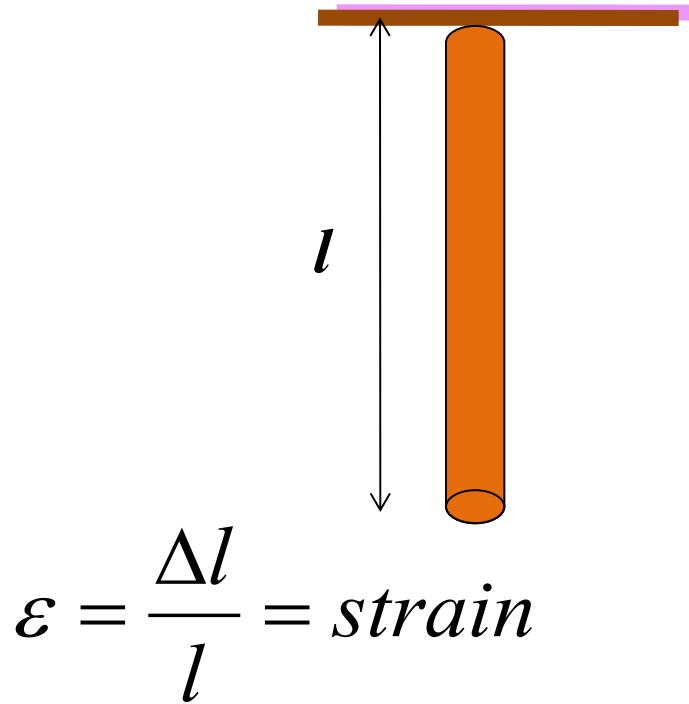


Earth's
composition

Ma Tu Vuliv A Pizza



Hook's Law



Hook's law

$$\mathbf{F = k * \Delta l}$$

F = Force

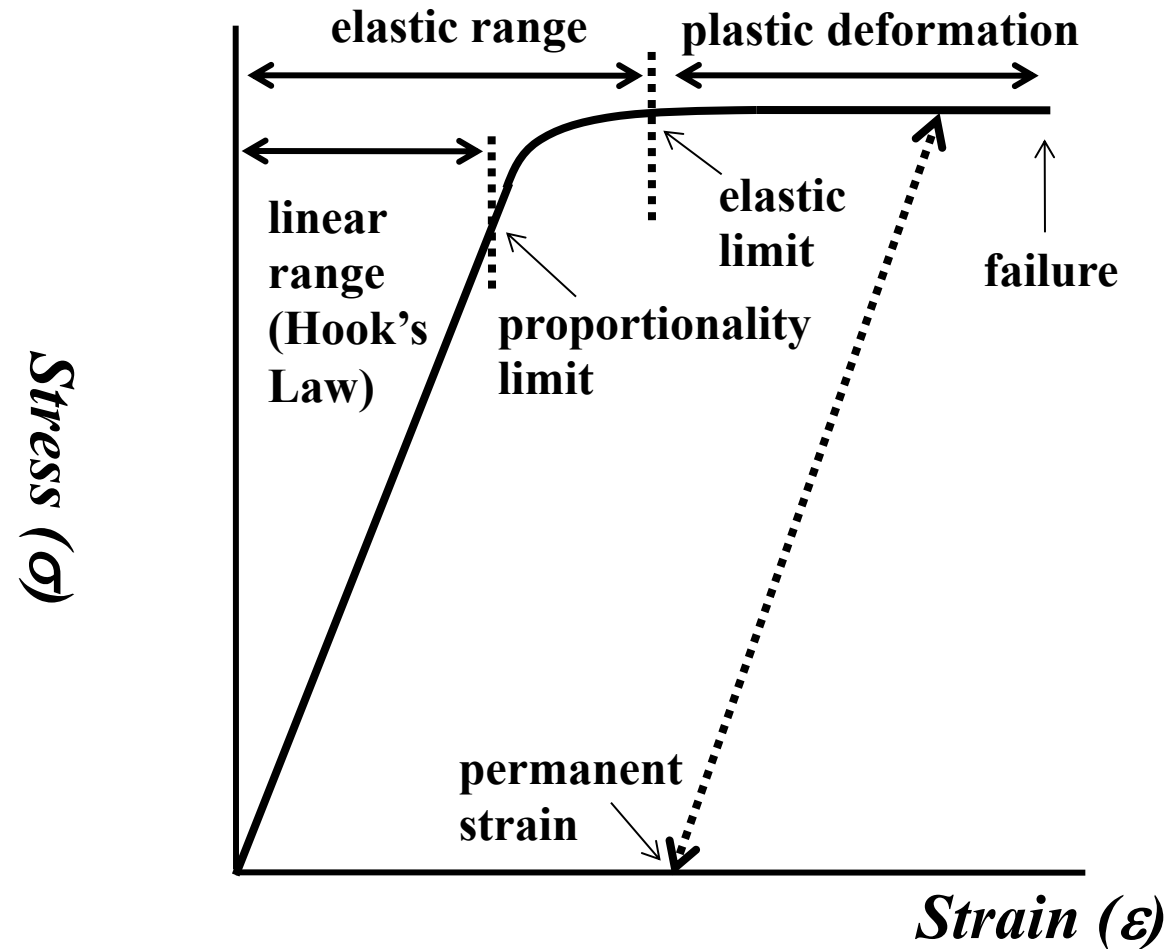
k = prop. const.

Δl = change in length

$$\Delta l \propto F l / A$$

$$\frac{F}{A} = \frac{\Delta l}{l}$$

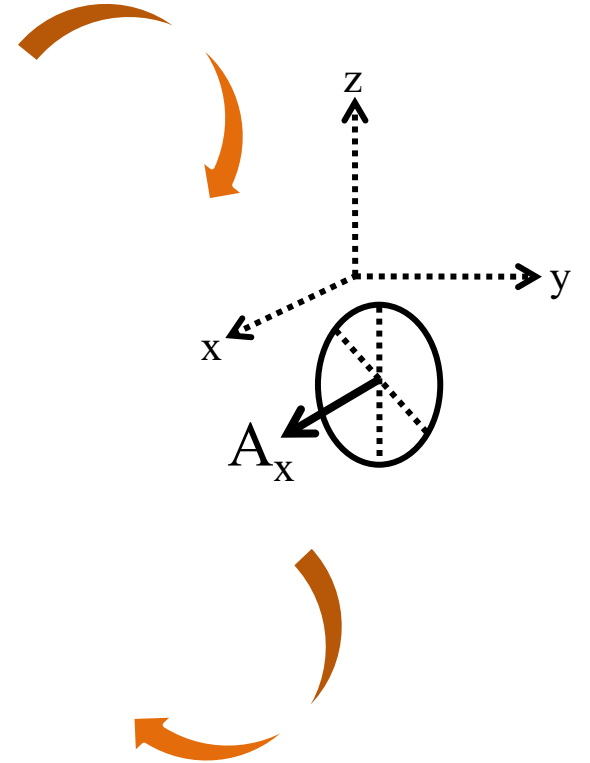
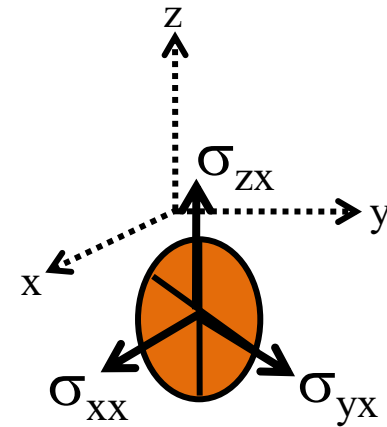
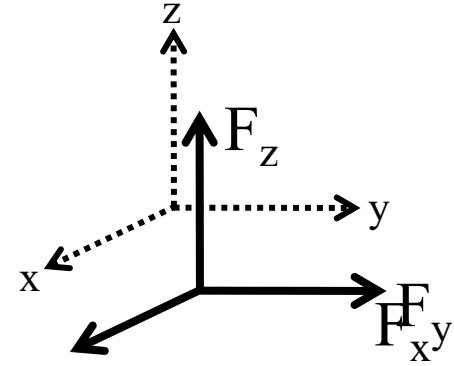
Elastic behaviour



To do calculations we need remain in the linear regime

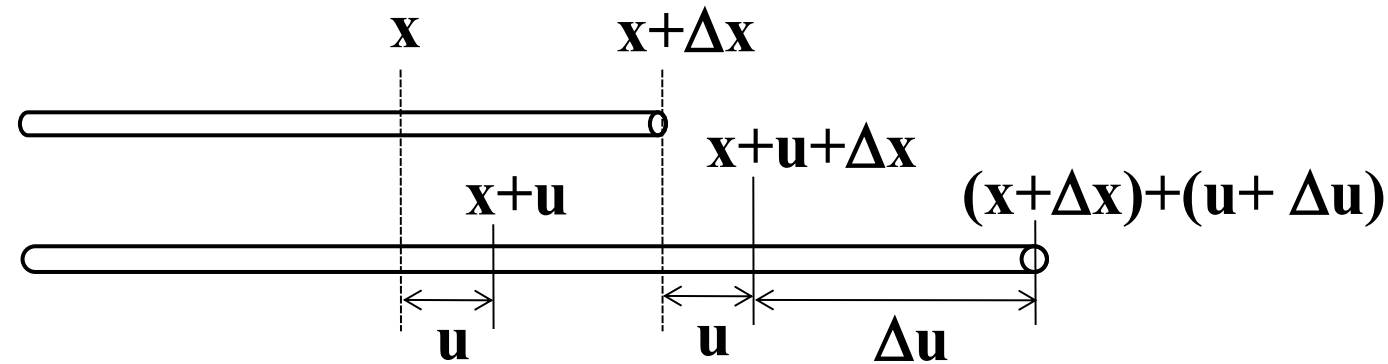
Stress Matrix

$$\sigma = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix}$$



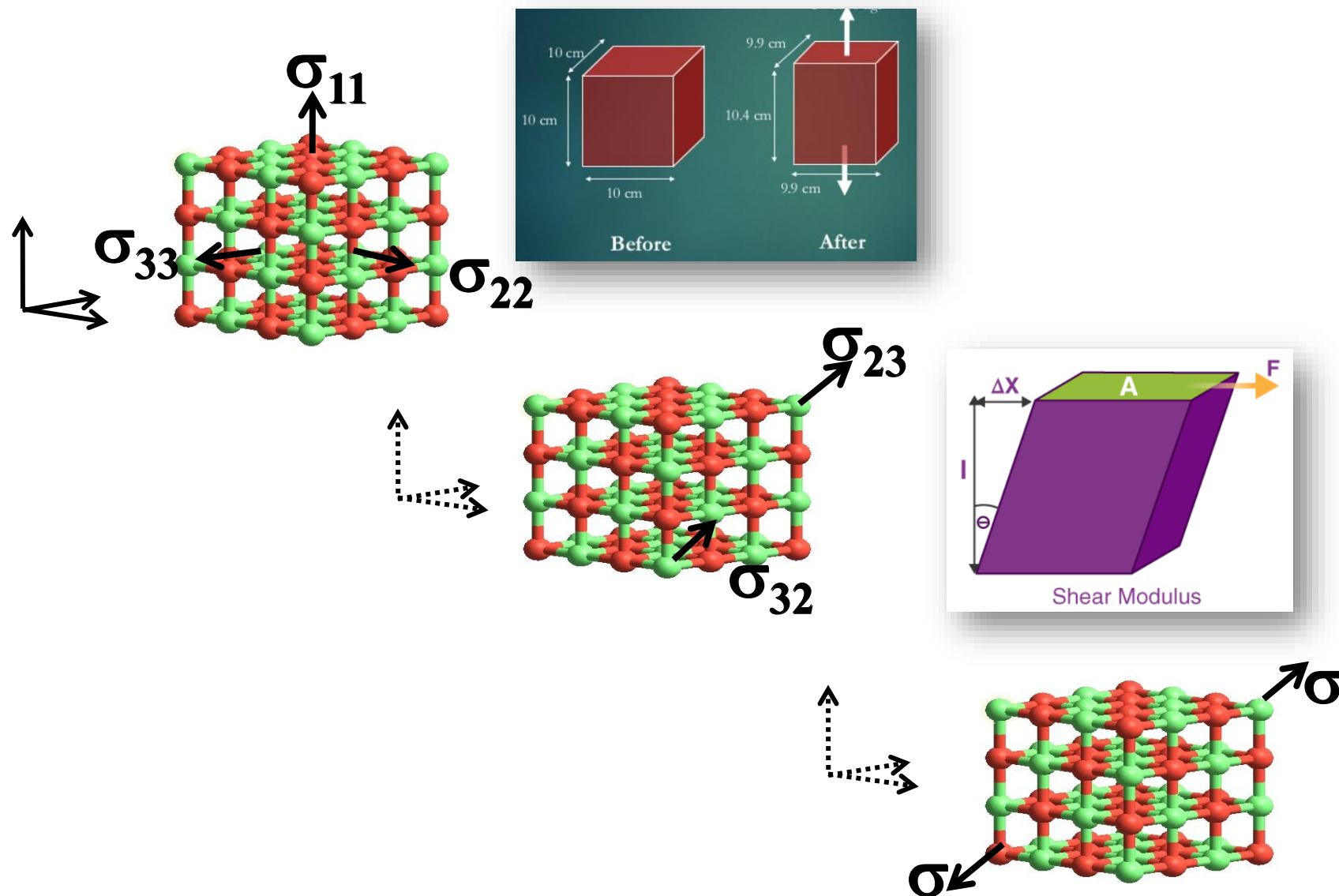
Strain Matrix

Strain involves a small displacement of point
 $r \rightarrow r'$



$$\varepsilon_{xx} = \frac{\text{change in separation}}{\text{original separation}} = \frac{\partial u}{\partial x}$$

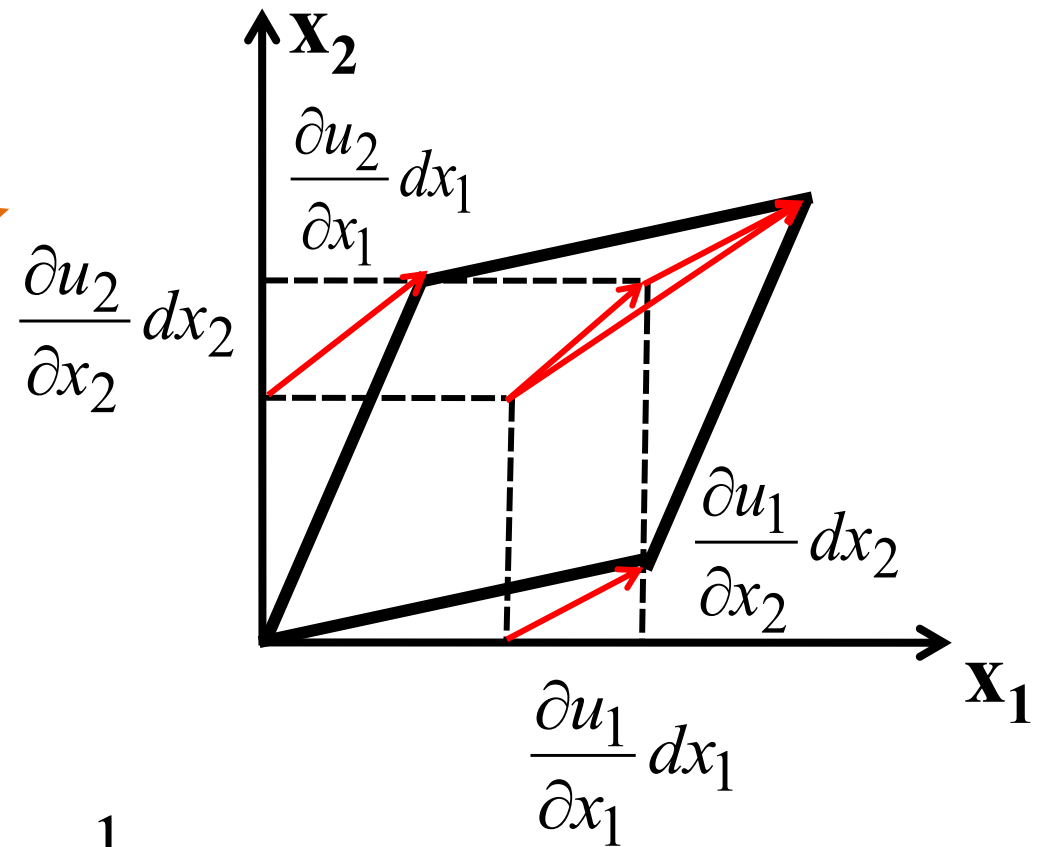
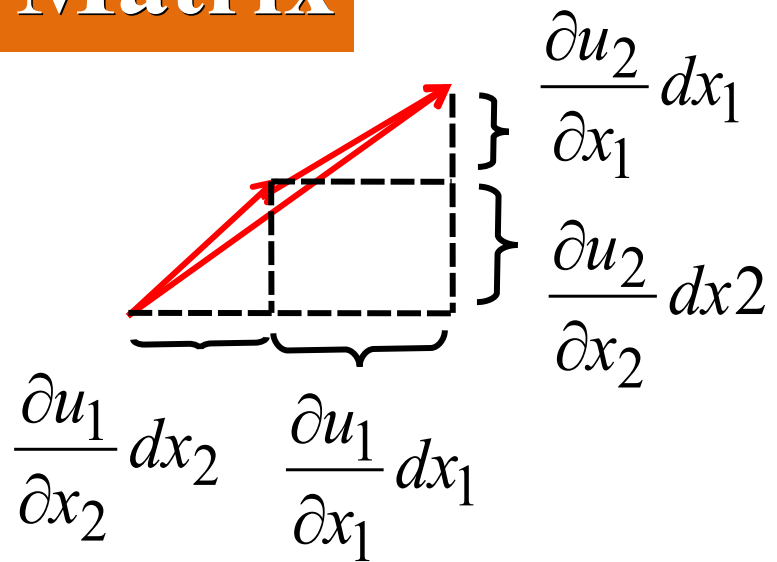
Three Distortions (strain displacements) in cubic case



Strain Matrix expressed by 3x3 matrix

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_{xx} & \varepsilon_{xy} & \varepsilon_{xz} \\ \varepsilon_{yx} & \varepsilon_{yy} & \varepsilon_{yz} \\ \varepsilon_{zx} & \varepsilon_{zy} & \varepsilon_{zz} \end{bmatrix}$$

Strain Matrix



Define the tensor of an infinitesimal strain ε_{ij} ($i, j=1, 2, 3$) as the symmetrical part of the displacement gradient tensor:

$$\varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

$$\varepsilon_{11} = \frac{1}{2} (\varepsilon_{11} + \varepsilon_{11})$$

$$\varepsilon_{12} = \frac{1}{2} (\varepsilon_{12} + \varepsilon_{21}) = \varepsilon_{21}$$

Elastic Tensor

The (Lagrangian) symmetric part of the Elastic tensor corresponds to pure strain and can be written as:

$$\varepsilon_{ij} = \frac{1}{2}(\varepsilon_{ij} + \varepsilon_{ji})$$



$$\varepsilon = \begin{pmatrix} \varepsilon_{11} & \frac{1}{2}(\varepsilon_{12} + \varepsilon_{21}) & \frac{1}{2}(\varepsilon_{13} + \varepsilon_{31}) \\ \frac{1}{2}(\varepsilon_{12} + \varepsilon_{21}) & \varepsilon_{22} & \frac{1}{2}(\varepsilon_{23} + \varepsilon_{32}) \\ \frac{1}{2}(\varepsilon_{13} + \varepsilon_{31}) & \frac{1}{2}(\varepsilon_{23} + \varepsilon_{32}) & \varepsilon_{33} \end{pmatrix}$$

Relation between stress and strain

For an isotropic, homogeneous solid and infinite small strains (*i.e.* Hook's law):

$$\sigma_{kl} = \sum_{mn} c_{klmn} \varepsilon_{mn}$$

The 4th order symmetrical tensor (c_{ijkl}) is the
ELASTIC CONSTANT TENSOR

**The elastic constants are usually expressed in contracted notations.
According to Voigt's notation:**

$$\begin{array}{ll} 11 \rightarrow 1 & 23=32 \rightarrow 4 \\ 22 \rightarrow 2 & 13=31 \rightarrow 5 \\ 33 \rightarrow 3 & 12=21 \rightarrow 6 \end{array}$$

$$c_{ij} = \frac{1}{V} \frac{\partial^2 E}{\partial \varepsilon_i \partial \varepsilon_j} \bigg|_0$$

where $i,j=1,6 \Rightarrow 6 \times 6$ matrix, which is symmetric $\Rightarrow$
 $\Rightarrow 21$ independent elements

Elastic Tensor

$$c_{ij} = \frac{1}{V} \frac{\partial^2 E}{\partial \varepsilon_i \partial \varepsilon_j} \bigg|_0$$

$$c_{ij} = \begin{pmatrix} c_{11} & c_{12} & c_{13} & c_{15} & c_{15} & c_{16} \\ c_{21} & c_{22} & c_{23} & c_{24} & c_{25} & c_{26} \\ c_{31} & c_{32} & c_{33} & c_{34} & c_{35} & c_{36} \\ c_{41} & c_{42} & c_{43} & c_{44} & c_{45} & c_{46} \\ c_{51} & c_{52} & c_{53} & c_{54} & c_{55} & c_{56} \\ c_{61} & c_{62} & c_{63} & c_{64} & c_{65} & c_{66} \end{pmatrix}$$

Taylor Expansion

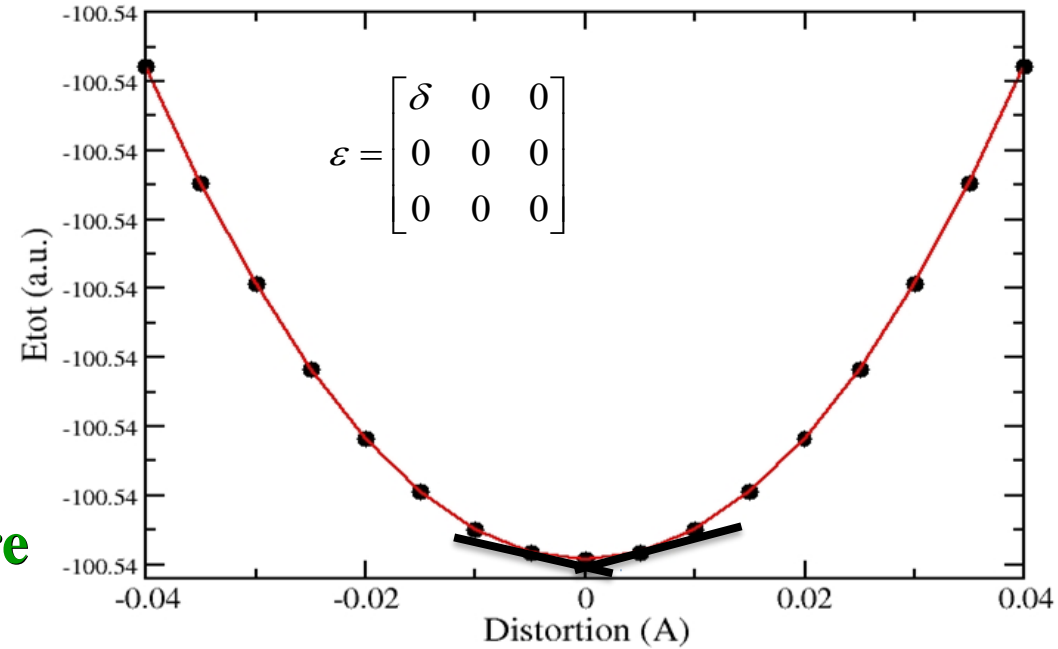
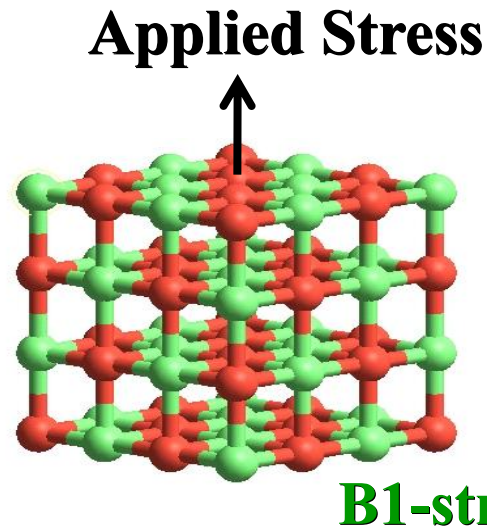
A Taylor expansion of the energy^{*)} to second order in the strain components gives:

$$E(\varepsilon) = E(0) + \underbrace{\sum_i^6 \frac{\partial E}{\partial \varepsilon_i} \bigg|_0}_{=0} \varepsilon_i + \frac{1}{2} \sum_{i,j}^6 \frac{\partial^2 E}{\partial \varepsilon_i \partial \varepsilon_j} \bigg|_0 \varepsilon_i \varepsilon_j$$

The elastic constants can be obtained by evaluating the energy as a function of strain.

^{*)} energy refer to the energy of the unit cell

ELASTIC CONSTANTS FROM 1ST-PRINCIPLES CALCUALTIONS



Calculate elastic constants using CRYSTAL:

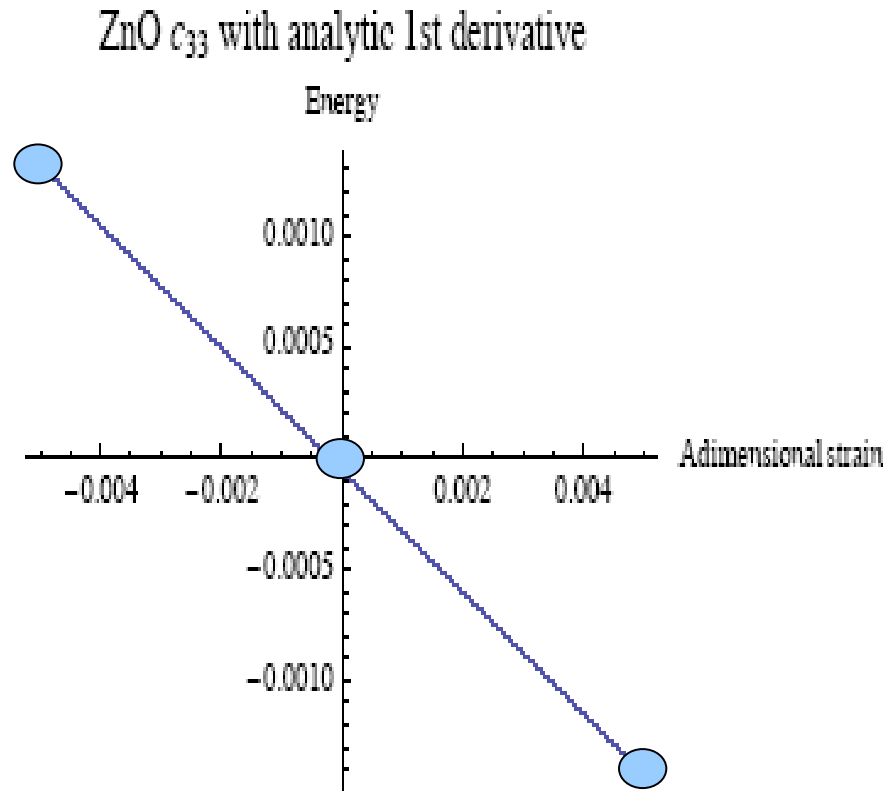
1. Optimise un-deformed structure
2. TENSOR (gives independent constants)
3. ELASTIC (deformation of unit cell)
4. OPTGEOM optimise the deformed structure if applicable)
5. From Energy potential curve determine “b” (by fitting)
6. Calculate elastic constants from “b”

ELASTIC CONSTANTS BY STRESS (automatic procedure) ELASTCON

See **CRYSTAL**
manual p. 288

If analytic first derivatives of the total energy are available, as in CRYSTAL, **then** fewer displacements are required because we need only fit to a linear function

- “**ELASTCON**” option
- **ELASTCON** option uses **NUMDERIV** points, default=3: $(-\delta, 0, \delta)$
- An **optimized input is assumed**, so the central-point calculation is simply to calculate the cell gradients at the equilibrium state

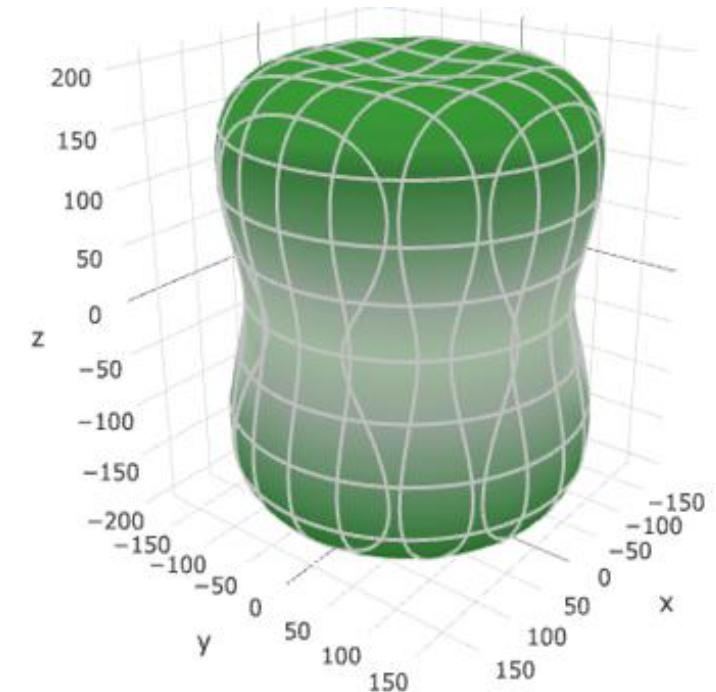


$$E(\delta) = -5.4 \times 10^{-5} - 0.27466\delta + \dots$$
$$b = 0.27466(9.90406) / 2 = 1.3601$$

USING ELASTCON OPTION IN CRYSTAL

Minimum of two keywords:

- **ELASTCON**
- **ENDELAST** (or just END)
- Part of Possible options:
- TOLDEG (or GTOL), RMS gradient tolerance; default = 6×10^{-5}
- TOLDEX (or XTOL), RMS displacement; default = 0.00012
- TOLDEE (or ETOL), Total energy convergence; default = 8
- **STEPSIZE**; default = 0.005
- **NUMDERIV**; default = 3
- **PRINT** (gives additional printing)
- **PREOPT**
- **RESTART**
- -----
- **PRESSEOS** – elastic constants under pressure
- **AWESOME** – seismic data



A. Erba

W.F. Perger, J. Criswell, B. Civalleri and R. Dovesi, Comp. Phys. Comm., 180, 1753-1759 (2009). Ab-initio calculation of elastic constants of crystalline systems with the CRYSTAL code

A. Erba, A. Mahmoud, R. Orlando and R. Dovesi, Phys. Chem. Minerals, 41, 151-160 (2014). Elastic properties of six silicate garnet end-members from accurate ab initio simulations.

Elastic constants for CUBIC case, in GPa

	C_{11}	C_{12}			
69.22	17.08	17.08	.	.	.
	69.22	17.08	.	.	.
		69.22	.	.	.
			58.25	.	.
				58.25	.
					58.25
			C_{44}		

ELASTIC CONSTANTS FOR ZnO

	ELASTCON	FROM ENERGY	Expt.
c_{11}	246.5	246.2	209.0
c_{12}	127.7	127.7	120.0
c_{13}	104.6	106.2	104.4
c_{33}	242.0	242.5	216.0
c_{44}	56.3	56.0	44.2
c_{66}	59.4	59.3	44.5

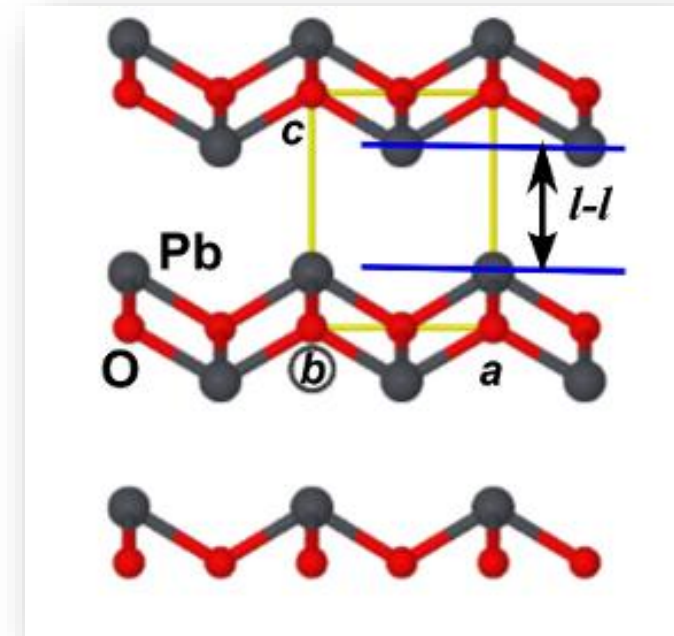
Elastic constants for α -PbO

Elastic constants in GPa

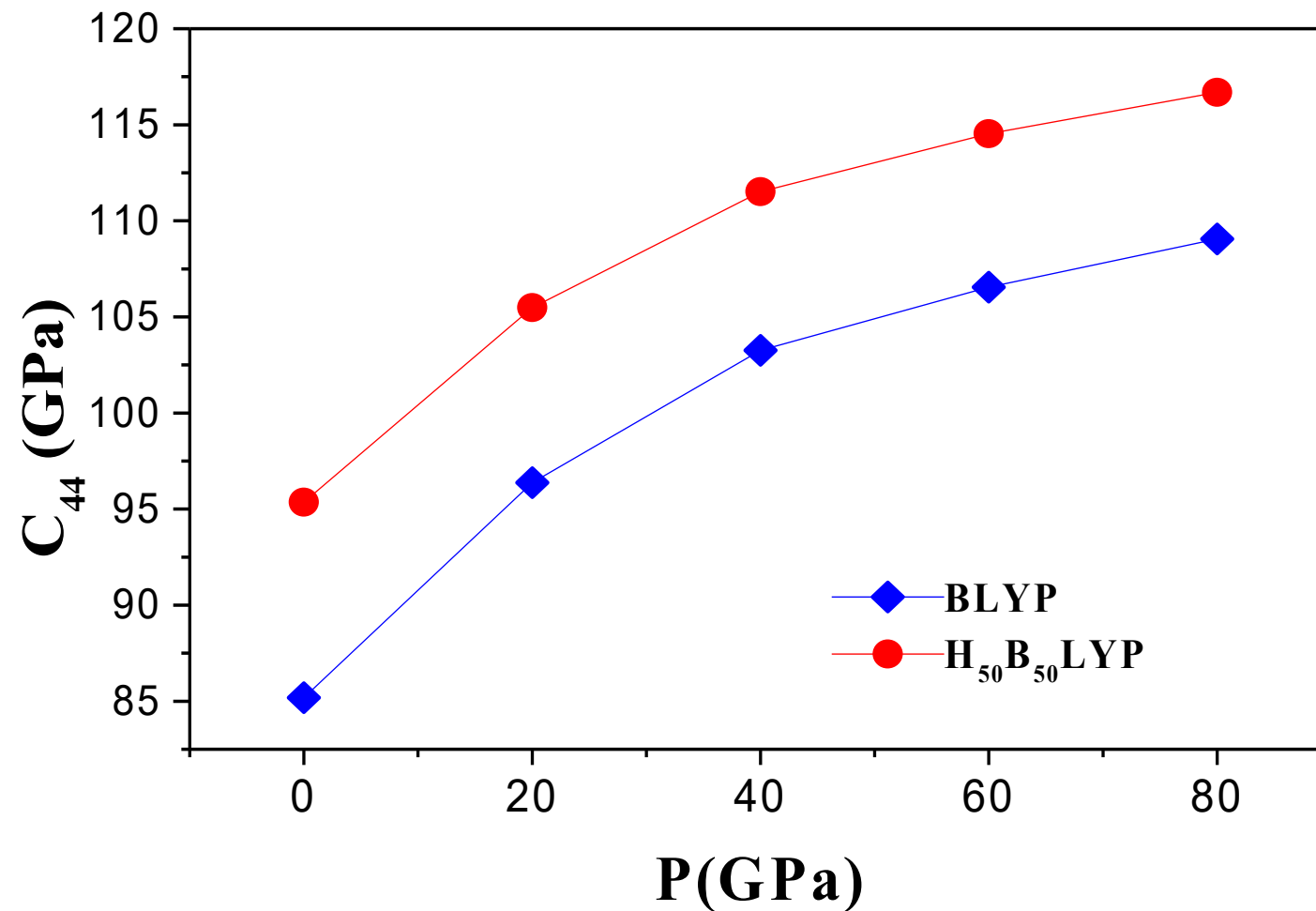
	C_{11}	C_{33}	C_{44}	C_{66}	C_{12}	C_{13}	$a(\text{\AA})$	$c(\text{\AA})$
PBE	65	16	11	55	65	15	4.10	5.00
DC	52	34	23	43	49	23	4.06	4.64

α -PbO

DC = Dispersion correction;
Grimme's description recalculated
 C_6^{ri} value.



Comparing Hamiltonians C44 for BaO



Elastic constants depend on the Hamiltonian

Properties using ELASTCON

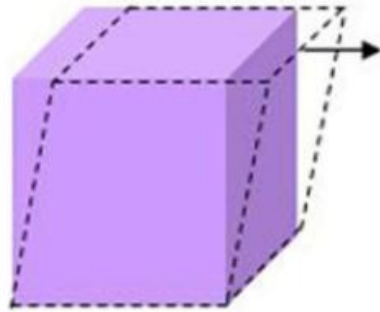
Easier:
use EOS
or ELASTCON

Young's
Modulus



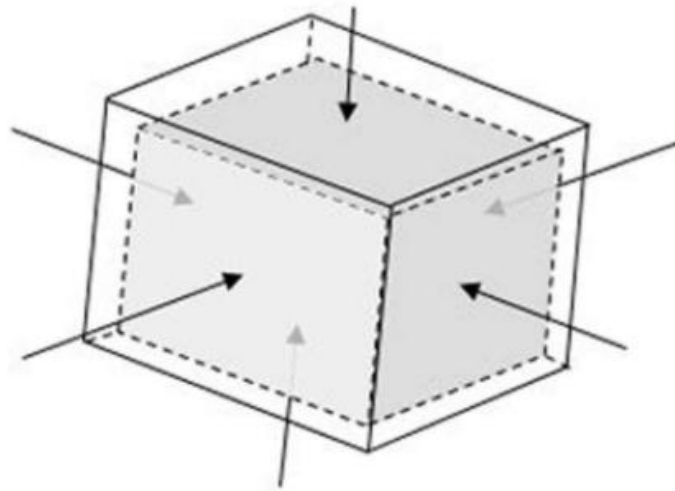
$$E = \frac{\sigma}{\epsilon}$$

Shear
Modulus



$$G = \frac{\tau}{\gamma}$$

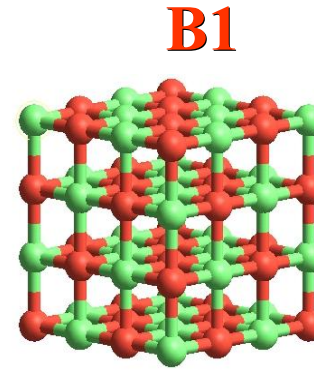
Bulk
Modulus



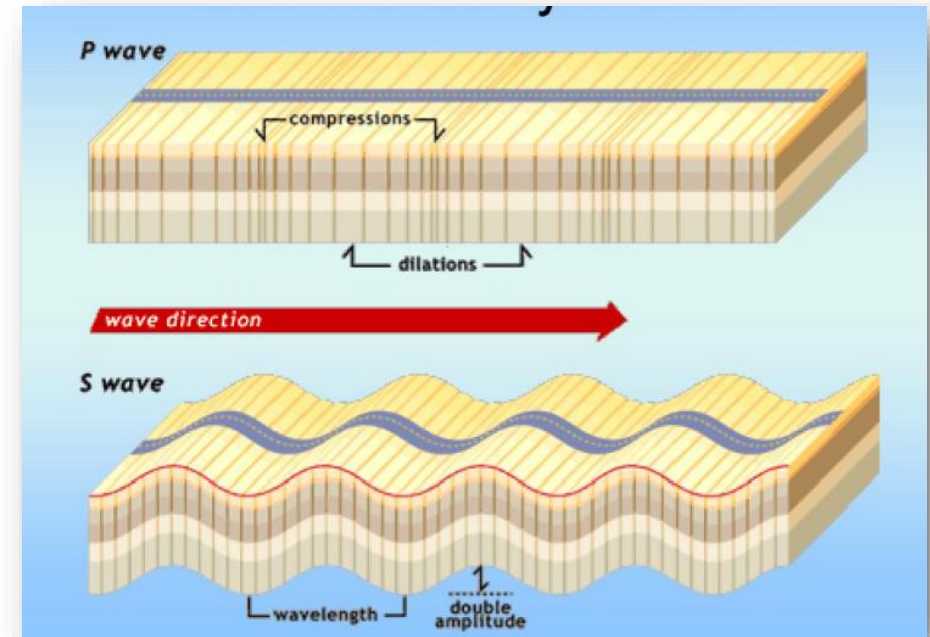
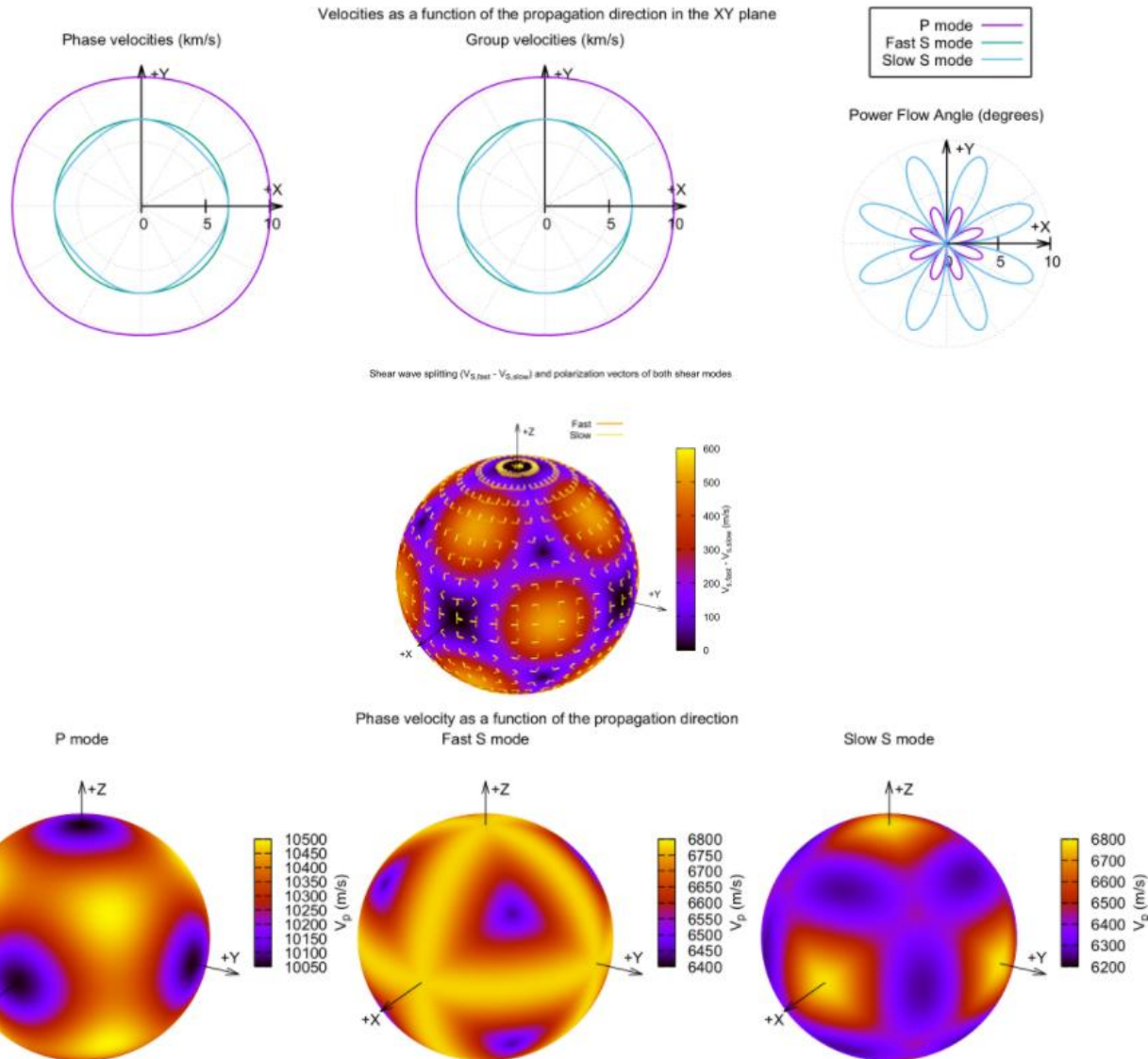
$$B = \frac{\sigma_{\text{hyd}}}{\Delta V/V_0}$$

- Bulk modulus
- Shear modulus
- Young's modulus
- Poisson ratio

Christoffel's equation



ELASTCON
AWESOME
END



Gnuplot scripts
available in Tutorial

Summary

Phase Transitions

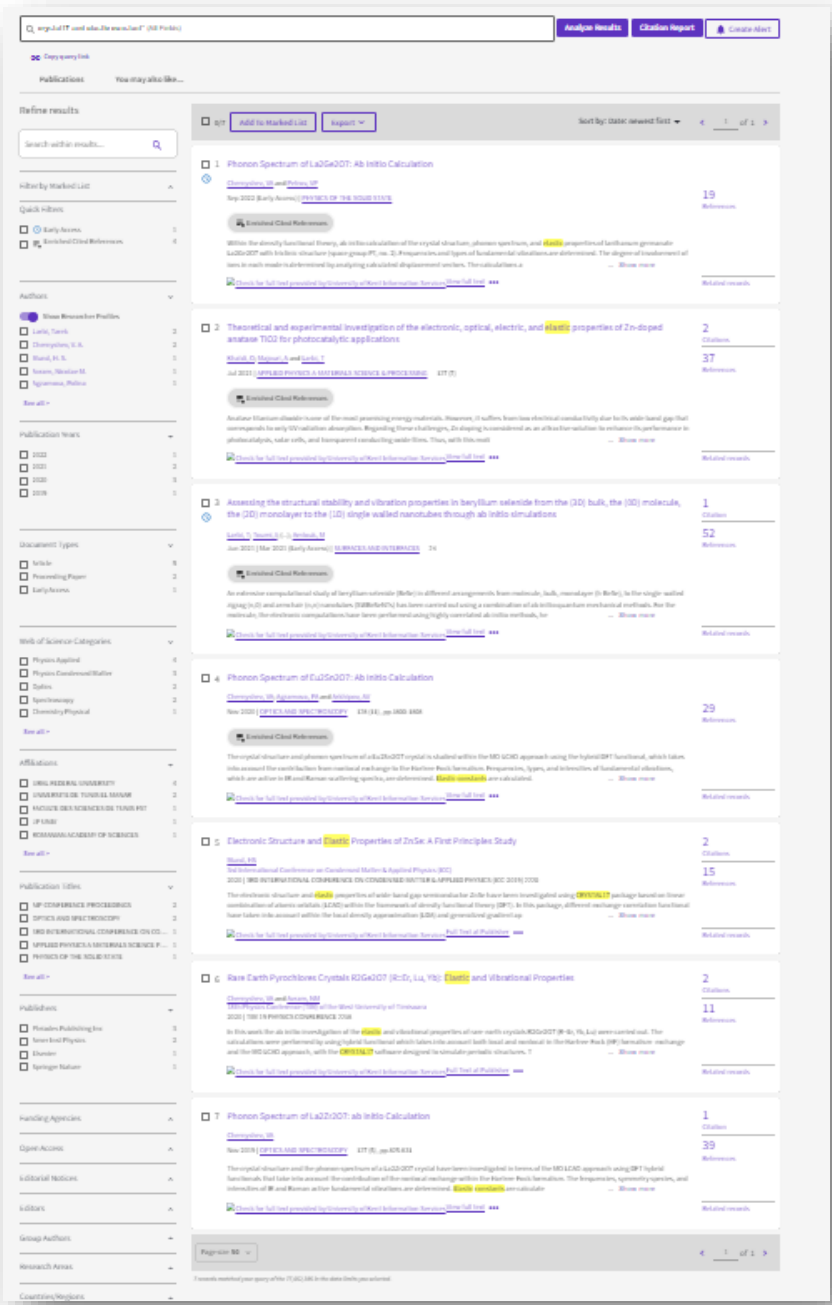
- Phase Transitions
- Pressure induced phase transitions

Equation of States (EOS)

- Bulk Modulus
- EOS for high-P
- Effect of Hamiltonian
- EOS

Elastic Constants

- From Energies
- ELASTCON



Tutorials available on:

- EOS and Elastic constants using EOS and ELASTCON
- Phase Transitions and Equation of States

Automated calculation of the elastic tensor of crystalline systems

- Generalized to 2D and 1D systems
- Calculation of directional seismic wave velocities
- Calculation of isotropic polycrystalline aggregates elastic properties via Voigt-Reuss-Hill scheme
- *Elastic Tensor under Pressure*
- *Complete Analysis of Elastic wave velocities through AWESOME Code*
- *Nuclear-relaxation Term through Internal-strain Tensor*

NEW in CRYSTAL23 – THERMO-ELASTIC CONSTANTS – p. 294 CR23 manual

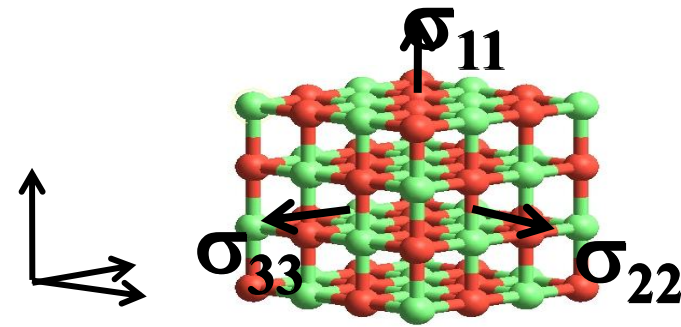
In Addition see Manual CRYSTAL23

Questions: m.l.alfredsson@kent.ac.uk

A SIMPLE (cubic) EXAMPLE

BULK MODULUS FROM ELASTIC CONSTANTS

$$\varepsilon = \begin{pmatrix} \delta & 0 & 0 \\ 0 & \delta & 0 \\ 0 & 0 & \delta \end{pmatrix} \longrightarrow B = \frac{(C_{11} + C_{12})}{3} \frac{2}{9V} b$$



- Bulk modulus
- Shear modulus
- Young's modulus
- Poisson ratio

Easier:
use EOS
or ELASTCON